Sol.

17. The earth's magnetic field is due to electric current produced by convection motion of metallic fluids (consisting mostly of molten iron and nickel) in the outer core of the earth. This is known as the dynamo effect.

18. At northern hemisphere earth's magnetic field is below the horizontal (angle of dip +ve) but at southern hemisphere earth's magnetic field above the horizontal (angle of dip –ve)

21. The declination in India is small, it being 0°41'E at Delhi and 0°58' W at Mumbai

22. A superconductor is perfectly diamagnetic material so, it moves from strong field to weak field.

23. Electrons in an atom orbiting around nucleus posses orbital angular momentum, these orbital electrons are equivalent to current carrying loop and thus possess orbital magnetic moment. The magnetic moment of single atom of dia magnetic material is zero. When magnetic field is applied, those electrons having orbital magnetic moment in the direction of field slow down and those in the opposite direction speed up. This happen due to induced current in accordance with Lenz's law.

24. When ferromagnetic substance is placed in external magnetic field, the size of domain lying in the direction of external magnetic field increases while size of domain lying in the opposite direction of field decreases.

25. Gauss theorem for electric field

$$\oint \vec{E}.d\vec{S} = \frac{q}{\varepsilon_0}$$

By comparing electric field and magnetic field

$$\vec{E} \equiv \vec{B},\ q \equiv q_m$$

$$\frac{1}{4\pi\varepsilon_0} \equiv \frac{\mu_0}{4\pi} \Rightarrow \frac{1}{\varepsilon_0} \equiv \mu_0$$

$$\boxed{\oint \vec{B}.d\vec{S} = \mu_0 q_m}$$

26. Magnetic lines form closed loops around a current carrying wire. The geometry of a straight solenoid is such a that magnetic lines cannot loop around circular wires without spilling over to the outside of solenoid .

The toroid have geometry such that magnetic lines can loop around electric wires without spilling over to the outside.

There is no force or torque on an element due to the field produced by that element itself

An atom of ferromagnetic and paramagnetic material have non-zero magnetic moment while their net charge is zero.

27. Jupiter has the fastest rotation rate and Venus has slower rate of rotation.

28. The magnetising current (I_m) is the additional current that needs to be passed through the windings of solenoid in the absence of the core which would give a magnetic field B value as in the presence of the core. Thus

$$B = \mu_0 n\ (I + I_m)$$

Here $B = \mu_0\mu_r nI$

$$\mu_0\mu_r nI = \mu_0 n\ (I + I_m) \Rightarrow I = 798A$$

29. The equatorial magnetic field of earth is

$$B = \frac{\mu_0 M}{4\pi r^3}$$

Here $B = 0.4\ G = 4 \times 10^{-5}$ T

r = radius of earth = 6.4×10^6 m

$$M = \frac{Br^3}{(\mu_0/4\pi)} = \frac{4\times10^{-5}\times(6.4\times10^6)^3}{10^{-7}}$$

$= 1.05 \times 10^{23}$ A-m^2

30. $M \propto \frac{B_{ext}}{T}$ for paramagnetic material

31. Definition of angle of declination is the angle between the geographic meridian and the magnetic meridian

32. For diamagnetic material χ_m is independent of temperature, for paramagnetic material

$\chi \propto \frac{1}{T}$, for ferromagnetic material.

$$\chi = \frac{C}{T - T_c} (T > T_c)$$

33. The hysteresis curve of large area has high coercivity & high coercivity which is used in permanent magnet, curve of small area has low coercivity & low remanence which is suitable for electromagnet & transfomer.

34. (1) The value of B at H = 0 is called remanence,
(2) Soft iron has low coercivity and low remanence, soft iron is suitable material for electromagnets
(3) The value of H at 'C' is called coercivity
(4) A permanent magnet has high remanence and high coercivity

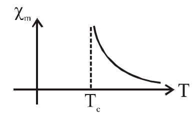
35. The field lines are completely expelled [for super conductors]
when $\chi = -1$ and $\mu_r = 0$

37. From graph at point 'b',
H = 0 so retentivity = 1.2T
at point 'c', B = 0 so coercivity = 90 A/m
at point 'a' saturation magnetic field = 1.5T

38. Susceptibility, for diamagnetic material is negative,
for paramagnetic material 'χ' is small positive,
for ferromagnetic material 'χ' is large positive

40. The value of vertical component of earth magnetic field at magnetic equator is zero.

42. Nickel is ferromagnetic material and curve between χ_m & T for this material is



When $T > T_c$ the curve represent the ferromagnetic be have like paramagnetic material.

43. (A) Wrong : magnetic field lines can never emanate from a point, over any closed surface, the net flux of B must always be zero.
(B) Wrong : magnetic field lines can never crosses each other
(C) Right : magnetic field lines completely confined within a Toroid.
Fig. (D) is wrong because field lines due to solenoid at it's ends and outside cannot be so completely straight and confined.

44. In southern hemi sphere angle of dip is negative & needle of dip circle becomes up. In Northern hemisphere angle of dip is positive & needle of dip circle becomes down.